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Effect of feeding a *Saccharomyces cerevisiae* fermentation product to Holstein cows exposed to high temperature and humidity conditions on milk production performance and efficiency—A pen-level trial

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ABSTRACT

The objective of this study was to evaluate the effect of feeding a *Saccharomyces cerevisiae* fermentation product (SCFP) on milk production efficiency of Holstein cows naturally exposed to high temperature and humidity conditions. The study was conducted in 2 commercial farms in Mexico from July to October 2020 and included 1 wk covariate period, 3 wk adaptation, and 12 wk data collection. Cows [n = 1,843; ≥21 d in milk (DIM) and <100 d carried calf] were enrolled and assigned to the study pens (n = 10) balanced for parity, milk yield, and DIM. Pens were fed a total mixed ration diet either without (CTRL) or with SCFP (19 g/d, NutriTek, Diamond V). Milk yield, energy-corrected milk (ECM), milk components, linear somatic cell score, dry matter intake (DMI), feed efficiency (FE; Milk/DMI and ECM/DMI), body condition score, and the incidence of clinical mastitis, pneumonia, and culling were monitored. Statistical analyses included mixed linear and logistic models accounting for repeated measures (when applicable; multiple measurements per cow within treated pens) with pen as the experimental unit and treatment, time (week of study), parity (1 vs. 2+), and their interactions as fixed and pen nested within farm and treatment as random effect. Parity 2+ cows within pens fed SCFP produced more milk than cows within CTRL pens (42.1 vs. 41.2 kg/d); there were no production differences between groups of primiparous groups. Cows within SCFP pens had lower DMI (25.2 vs. 26.0 kg/d) and greater FE (1.59 vs. 1.53) and ECM FE (1.73 vs. 1.68) than cows within CTRL pens. Milk components, linear somatic cell score, health events, and culling were not different between groups. At the end of the study (245 ± 54 DIM), SCFP cows had greater body condition score than CTRL (3.33 vs. 3.23 in the first parity; 3.11 vs. 3.04 in 2+ parity cows). Feeding *Saccharomyces cerevisiae* fermentation products to lactating cows exposed to high temperature and humidity conditions improved FE.

Key words: postbiotic, heat stress, feed efficiency, body condition score

INTRODUCTION

Heat stress (**HS**) and its effect on immunity, gut integrity, welfare, and production efficiency are major concerns for the dairy industry (Polsky and von Keyserlingk, 2017; Becker et al., 2020). Cooling strategies, genetic selection, and nutritional management are examples of on-farm management practices to mitigate HS and its consequences (Becker et al., 2020). Postbiotic (inanimate microorganisms and their components; Salminen et al., 2021) products of *Saccharomyces cerevisiae* fermentation (**SCFP**) are among the nutritional management strategies available. Its anti-inflammatory properties (Jensen et al., 2008; Knoblock et al., 2019), as well as the ability to modulate immune responses (Mahmoud et al., 2020) and influence rumen pH (Thrune et al., 2009) and microbiota (Tun et al., 2020), may present additional value in mitigating the effects of HS in dairy cattle.

Heat stress occurs when cattle are exposed to a high temperature, humidity, or a combination of both, leading to the inability to dissipate heat produced through normal metabolism (Dahl et al., 2020). Heat stress and inflammation are associated with disruption of normal metabolism and poor performance of dairy cows (McCarthy et al., 2016; Becker et al., 2020). For example, cows exposed to severe HS reduced their DMI by approximately 50% and milk yield by 27% (Al-Qaisi et al., 2020a) and were at greater risk of health disorders (e.g., retained placenta, metritis, mastitis) than cows exposed to thermoneutral temperature-humidity index (**THI**; Menta et al., 2022). The physiologic mechanism between HS and detrimental performance effects is considered complex and multifactorial (Bradford et al.,

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2015). The uncontrolled inflammatory process secondary to elevated THI exposure, as well as the increased energy demand toward body cooling and reduction in DMI appear to be the primary factors responsible for lower milk yield, more disease events, and decreased reproductive performance (Burhans et al., 2022). Gastrointestinal permeability secondary to HS has been suggested as a major factor to a greater magnitude of systemic inflammation (Sanz-Fernandez et al., 2014; Burhans et al., 2022; Al-Qaisi et al., 2020a), which may partially explain the greater incidence of disease observed under HS conditions (Menta et al., 2022).

Saccharomyces cerevisiae fermentation products have previously demonstrated advantages in controlling inflammation and improving health and performance of dairy cows. For example, Knoblock et al. (2019) demonstrated that transition cows (n = 38) fed rations containing SCFP had reduced serum haptoglobin concentrations compared with control cows at 7 d postpartum (0.6 vs. 0.25 mg/mL, respectively). In mid-lactation dairy cows, Acharya et al. (2017) concluded that dietary supplementation with SCFP (n = 80 cows) was associated with increased milk yield (+3 kg/d) and feed efficiency (**FE**; 1.41 vs. 1.30 kg/kg). Although a cause-effect relationship has not been established, it seems plausible that by controlling inflammatory responses and improving rumen microbiota and their function, dairy cows will have greater FE. Previous research performed under HS condition have demonstrated anti-inflammatory properties (lower blood serum amyloid A and cortisol concentrations; Al-Qaisi et al., 2020b), improved net energy balance (11% improvement; Zhu et al., 2016), and improved FE (7% improvement; Schingoethe et al., 2004) in cows fed SCFP supplements.

To our knowledge, the effects of SCFP fed to lactating cows exposed to HS conditions on multiple commercial dairy farms have not been studied. The main objectives of this randomized controlled trial were to evaluate the effect of feeding a SCFP to lactating Holstein cows exposed to THI danger zone conditions on milk yield and FE in commercial dairy herds in Mexico. We hypothesized that in cows naturally exposed to HS conditions, the group fed SCFP would have greater FE (increased milk yield, decreased DMI, or both) than cows exposed to the same environmental conditions but not fed SCFP.

MATERIALS AND METHODS

This work was designed and written considering the reporting guidelines for randomized controlled trials (Sargeant et al., 2010), and conducted in accordance with the principles and specific guidelines presented in Guide for the Care and Use of Agricultural Animals in Research and Teaching (McGlone et al., 2010).

Animals, Facilities, and Management

This pen-level randomized controlled trial was conducted on 2 commercial dairy farms (farm A and B) located in Torreón, Coahuila, Mexico, from July until October 2020. Farm A had approximately 6,000 milking cows, with overall average milk production during summer months of 36 kg/cow per day. Farm B had approximately 2,300 milking cows and an average milk production during summer months of 35 kg/cow per day. In this region, summer months are characterized by high environmental temperatures and temperature-humidity index (**THI**). During the study, THI was continuously recorded for descriptive purposes (Figure 1). The number of hours per day exposed to THI ≥68 during the data collection period is presented in Tables 1, 2, 3, 4, 5.

Holstein cows (n = 1,843; farm A = 1,093, farm B = 750) without clinical signs of disease, between 21 and 180 DIM, were randomly assigned to each pen. Within each pair of pens (5 pairs; Tables 1 and 4), a coin was flipped to assign one of 2 dietary treatments. All cow data were retrieved from each farm management software before starting date and randomization of cows to pens was performed using RAND function in Excel (Microsoft Corp.). Within a pair of pens, allocation of cows was balanced considering lactation number, DIM, average daily milk production, and clinical mastitis events (within 7 d before start of the study; Table 4). Pregnant cows had to be less than 100 d carried calf to ensure completion of the study before their dry-off date. After initial randomization, no cows were moved in or out of the allocated pen (closed population) unless they needed to be treated, were marketed or died.

Cows were housed in shaded dry-lot pens with covered feeding alleys and were milked 3 times per day. Fans and misters or sprinklers were used as cooling strategies in the parlor holding area and in cooling pens. Cows were moved to cooling pens and exposed to sprinklers and fans 6 times daily for 30 min per session (upon exit of the milking parlor and between each milking session). Cooling pens are part of the HS mitigation strategies applied on farms located in areas exposed to extremely high environmental temperature in Mexico. Cooling pens are on-farm designated areas where cows are routinely and temporarily moved into for the only purpose of cooling. Cows had free access to water before entering the holding pen and after exiting the milking parlor into the cooling pens.

Experimental Design and Treatment Groups

The experiment followed a complete randomized design. Within farm, multiple pairs of similar pens were enrolled in the study as paired groups (Table 1). Each

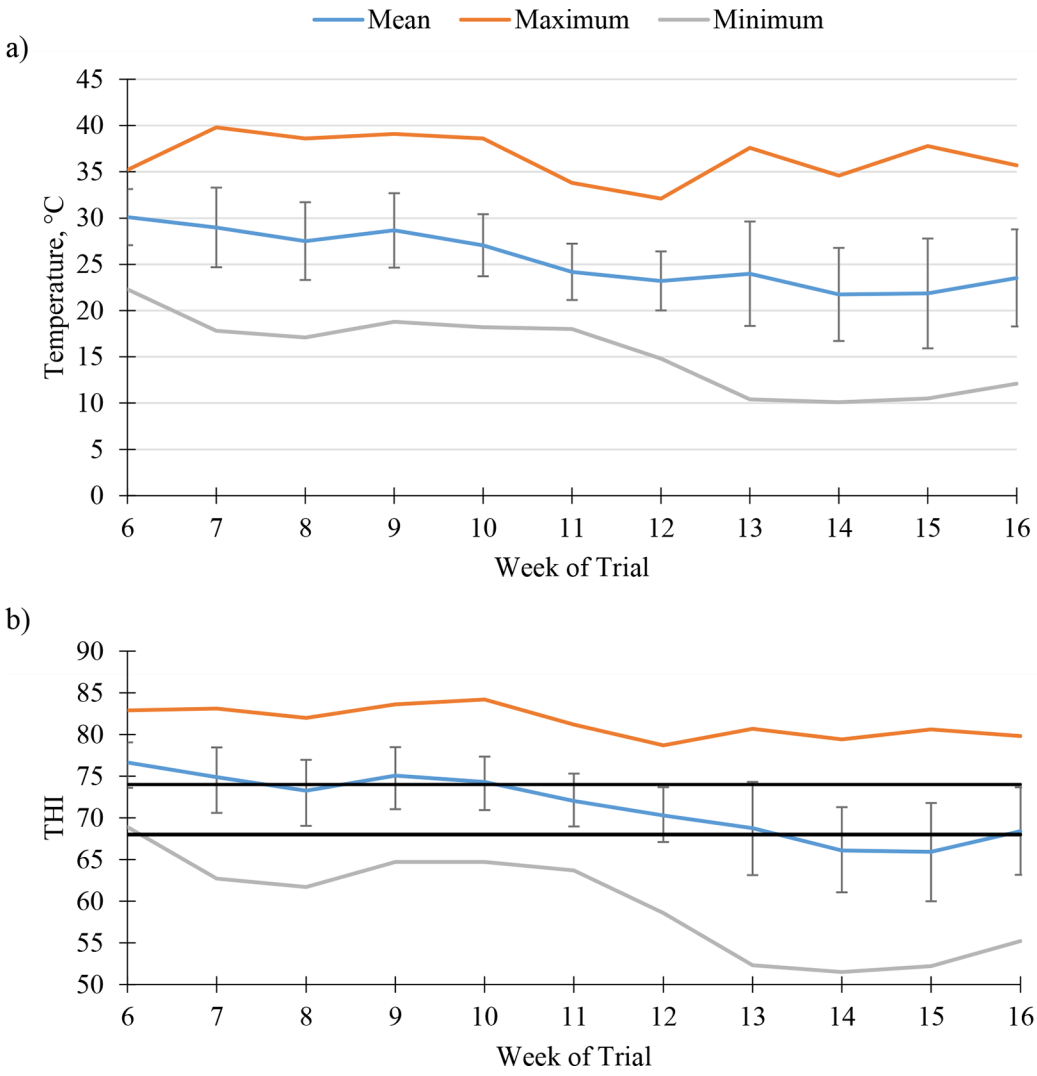


Figure 1. Descriptive statistics of weekly (a) environmental temperature and (b) temperature-humidity index (THI) at 2 farms enrolled in a pen randomized controlled trial throughout the data collection period. Error bars represent the standard deviation; horizontal bold black lines represent the 68 and 74 THI thresholds.

Table 1. Representation of study design and experimental units[1] through the study period (wk 1–16) of a pen-level controlled trial of the effect of supplementing lactating TMR with *Saccharomyces cerevisiae* fermentation product (SCFP) or without supplementation (CTRL) on 2 commercial dairy farms

	Farm A						Farm B			
	Pair 1: primiparous cows		Pair 2: multiparous cows		Pair 3: multiparous cows		Pair 4: mixed parities		Pair 5: mixed parities	
Week of trial	Pen 1	Pen 2	Pen 3	Pen 4	Pen 5	Pen 6	Pen 7	Pen 8	Pen 9	Pen 10
1 (covariate period)	CTRL	CTRL	CTRL	CTRL	CTRL	CTRL	CTRL	CTRL	CTRL	CTRL
2 to 4 (adaptation)	CTRL	SCFP	SCFP	CTRL	CTRL	SCFP	CTRL	SCFP	CTRL	SCFP
5 to 16 (data collection)	CTRL	SCFP	SCFP	CTRL	CTRL	SCFP	CTRL	SCFP	CTRL	SCFP

[1]5 CTRL pens; 5 SCFP pens.

Table 2. Ingredient composition and nutrient profile of lactating diets fed in a pen-level controlled trial to assess the effects of a *Saccharomyces cerevisiae* fermentation product (CTRL diet: TMR without supplement; SCFP diet: TMR with a *Saccharomyces cerevisiae* supplement) on milk production efficiency during heat stress conditions[1]

Ingredient, % of DM	Farm A	Farm B
Alfalfa hay	9.4	13.2
Sorghum silage		1.9
Corn silage	31.5	24.1
Soybean hulls	2.6	
Cottonseed	7.3	6.5
Steam-flaked corn	25.6	32.4
Molasses	1.2	2.4
Sugarbeet pellet	2.3	
Soybean meal	13.0	10.7
Canola meal	1.6	4.8
Vitamin mineral mix[2]	2.2	2.4
Sodium bicarbonate	0.2	
High-palmitic acid fat supplement[3]	1.3	0.6
Calcium salts of fatty acids supplement[4]	1.2	1.0
Heat-treated soy[5]	0.5	
Methionine analog[6]	0.1	0.1
CTRL or SCFP supplement[7]	0.1	0.1

[1]This pen-level randomized controlled study was performed simultaneously on 2 commercial dairy farms.

[2]Vitamin-mineral mix contained the following ingredients: 35% sodium bicarbonate, 32% calcium carbonate, 13.05% salt white, 10% magnesium oxide, calcium phosphate 4.3%, wheat mill run 2.7%, mineral oil 1.0%, Zinpro Availa Dairy 0.8% (Zinpro Corp.), zinc sulfate, 0.36%, Rumensin 0.3% (Elanco Animal Health), vitamin E, 0.2% (BASF Corp.), manganase sulfate 0.17%, copper sulfate, 0.01%, EDDI 0.01%, sodium selenite, 0.09%, vitamin A 0.014% (BASF Corp.), and vitamin D 0.009% (DSM Nutritional Products).

[3]Commercial fat supplement contained a minimum of 85% palmitic acid; Jefo Nutrition Inc.

[4]Commercial fat supplement contained calcium salts of fatty acids; Lactomil, Industrializadora Oleofinos S.A. de C.V.

[5]Expeller soybean meal, Soy 70, Nutrilag).

[6]DL-Methionine hydroxy analogue calcium, Novus International Inc.

[7]On farm A, a placebo was included in the CTRL diet (rice hulls) and a *Saccharomyces cerevisiae* fermentation product was included in the SCFP diet (NutriTek, DiamondV). On farm B, a placebo was not used. The *Saccharomyces cerevisiae* fermentation product was added to the TMR mixer after TMR was delivered to the CTRL pens.

pair of pens had the same stall layout and size, were orientated similarly relative to the milking facility, and provided the same access to lying area, shade, bunk space, water, and heat abatement.

Pens of a paired group were assigned to one of 2 diets: control (**CTRL**; lactating TMR without SCFP) or SCFP treatment diet (lactating TMR supplemented with SCFP, NutriTek, Diamond V, Cedar Rapids, IA). The nutrient profile of the diets was the same for CTRL and SCFP pens within each farm (Table 3). A total of 5 pairs of pens were enrolled (10 pens: 5 CTRL and 5 SCFP). Three pairs of pens were enrolled in farm A (2 pairs of multiparous cows and 1 pair of primiparous cows) and 2 pairs in farm B [mixed parity pens; primiparous (49%) and multiparous cows commingling].

After initial cow randomization, all pens received the same diet (CTRL) for one week to obtain covariate information (baseline data; Table 6). After this period, only one diet was offered to each pen (CTRL or SCFP diets). A 3-wk period was provided to allow for adaptation to assigned treatment lactating diets. Observations (12 wk) were collected at the beginning of wk 5 until the end of the study period. The total duration of the study was 16 wk, including covariate period, adaptation to treatment diets, and data collection periods (Table 1). Farm and research staff were not blind to dietary treatments.

The primary outcomes of interests were milk yield, ECM, DMI, FE, and BCS. Additionally, we assessed the treatment effect on the incidence of clinical mastitis, pneumonia, and culling (sold and dead). Clinical mastitis events (visible abnormal milk) were diagnosed during the milking routine and pneumonia cases were diagnosed by on-farm trained personnel, defined as elevated respiratory effort with fever (>39.5°C), with or without nasal or ocular discharge or cough.

Experimental treatments were applied at the pen-level; therefore, pen was considered the experimental unit. Depending on the outcome of interest analyzed, individual cows within the pens or the pen were considered observational units (St-Pierre, 2007; Tempelman, 2009; Bello et al., 2016). Sample size (i.e., number of pens per treatment) was calculated using the method described in Tempelman (2009). Briefly, we estimated the number of pens needed to detect a difference in milk production of at least 2 kg/cow per day (Acharya et al., 2017), using 140 cows per pen, a within-cow variability of 3 kg^2, a between-cow within treatment variability of 3 kg^2, and a between pens within treatment variability of 1 kg^2 (Tempelman, 2009). Based on Equation [1], the estimated effect size (Δ) for that difference in milk production was 1.96 and rounded to 2.0 to match the values presented for power and sample size curves provided in Tempelman (2009).

$$\Delta = \frac{\delta}{\sqrt{\sigma_r^2}}, \quad [1]$$

where

$$\sigma_r^2 = \frac{\sigma_e^2 + \sigma_{c(t)}^2}{d} + \sigma_{p(t)}^2,$$

and δ is the expected mean difference between groups and σ_r^2 is the composite variance, which was calculated based on the within-cow variability (σ_e^2), the between-cow within treatment variability $\left(\sigma_{c(t)}^2\right)$, the between

Table 3. Nutrient profile of lactating diets fed in a pen-level controlled trial to assess the effects of a *Saccharomyces cerevisiae* fermentation product (CTRL diet: TMR without supplement; SCFP diet: TMR with the supplement) on milk production efficiency[1]

	Farm A		Farm B	
Nutrient,[2] DM basis	CTRL	SCFP	CTRL	SCFP
DM, % as fed	49.8 ± 1.3	50.2 ± 1.4	52.2 ± 1.01	52.0 ± 1.0
CP, % DM	17.4 ± 0.6	17.5 ± 0.7	17.4 ± 0.5	17.4 ± 0.3
Soluble protein, %	32.55 ± 4.67	33.43 ± 4.57	34.1 ± 2.57	33.96 ± 2.89
ADF, % DM	19.03 ± 1.29	19.29 ± 1.06	18.6 ± 0.93	18.43 ± 0.87
aNDFom, % DM	26.21 ± 1.59	25.93 ± 1.33	26.23 ± 1.2	24.93 ± 0.94
Lignin, % DM	3.36 ± 0.29	3.4 ± 0.51	3.7 ± 0.17	3.68 ± 0.37
Starch, % DM	23.76 ± 1.67	23.25 ± 1.69	23.96 ± 2.25	24.48 ± 1.13
Sugar WSC, % DM	5.29 ± 1.14	5.37 ± 1.16	6.66 ± 1.53	6.57 ± 1.69
Sugar ESC, % DM	4.56 ± 1.2	4.63 ± 1.25	5.66 ± 1.13	5.5 ± 1.53
Fat EE, % DM	5.02 ± 0.3	5.13 ± 0.24	4.82 ± 0.86	4.79 ± 0.7
Ash, % DM	8.63 ± 0.3	8.67 ± 0.22	8.49 ± 0.47	8.5 ± 0.67

[1]This study was performed simultaneously on 2 commercial dairy farms. Values are presented as mean ± SD. Values obtained by near infrared reflectance spectroscopy analysis of TMR samples. The TMR samples were collected weekly for each pen. Monthly composites by pen were submitted to a commercial laboratory (Rock River, Torreón, Mexico) for analysis of nutrient profile.

[2]aNDFom = amylase neutral detergent fiber organic matter; WSC = water-soluble carbohydrates; ESC = ethanol-soluble carbohydrate; EE = ether extract.

pens within treatment variability $\left(\sigma^2_{p(t)}\right)$, and the number of animals per pen (d). Based on the sample size curves provided in Tempelman (2009) for completely randomized designs, 5 pens per treatment were required to achieve 80% power with α of 5% for an effect size (Δ) of 2.00 with the number of cows available per pen to measure milk production.

Total Mixed Ration Formulations, Preparation, and Delivery

The lactating diet was formulated using NDS Professional version 3.9.9.07 (RUM&N Sas) based on Cornell Net Carbohydrate and Protein System guidelines (version 6.5.5; Van Amburgh et al., 2015) to meet postpartum requirements and the milk yield objectives of the farms. In both farms, TMR was delivered to cows 6 times daily as part of the farm's standard feeding protocols and daily schedule considering approximately 3% refusals. Feeding schedule was managed in a commercial feed management software (TMR Tracker, Digi-Star, Topcon Positioning Group). Cows had access to ad libitum water.

In the SCFP diet, a Microtracer RF (Micro-Tracers Inc.) was added at a rate of 10 g/t. Evaluation of the tracers within the final TMR offered to the respective pens confirmed the inclusion of the appropriate supplement in the diet. This tracking was completed twice a week on both farms at random days throughout the study.

Table 4. Description of the groups by farm, replicate, and pen enrolled in a pen-level controlled trial to assess the effects of a *Saccharomyces cerevisiae* fermentation product (CTRL diet: TMR without supplement; SCFP diet: TMR with the supplement) on milk production efficiency[1]

Farm	Replicate[2]	Pen	Treatment group	n cows	n lactation	DIM	Milk yield[3] (kg/cow per d)
A	1	1	CTRL	275	1.0 ± 0.0	155 ± 21	37.2 ± 5.3
		2	SCFP	271	1.0 ± 0.0	155 ± 21	36.9 ± 5.1
	2	3	SCFP	137	3.2 ± 1.0	110 ± 46	48.9 ± 8.3
		4	CTRL	137	3.2 ± 1.0	113 ± 45	49.4 ± 7.5
	3	5	CTRL	137	3.2 ± 1.0	111 ± 45	48.1 ± 8.1
		6	SCFP	136	3.1 ± 0.9	110 ± 45	47.4 ± 7.5
B	4	7	CTRL	188	1.9 ± 1.2	133 ± 72	39.8 ± 6.0
		8	SCFP	189	2.0 ± 1.3	131 ± 68	39.6 ± 6.0
	5	9	CTRL	186	1.9 ± 1.1	131 ± 69	40.4 ± 6.1
		10	SCFP	187	1.8 ± 1.1	134 ± 71	39.8 ± 5.3

[1]Values are presented as mean ± SD.

[2]Replicate 1 had primiparous cows only. Replicates 2 and 3 had only multiparous cows, whereas replicates 4 and 5 included mixed parity groups.

[3]Values obtained during wk 1 of the study period (covariate period).

Table 5. Descriptive statistics [minimum (Min), mean, and maximum (Max)] of the number of hours per day spent above temperature-humidity index (THI) thresholds of 68 and 74 by month and week of trial (WOT) on 2 farms enrolled in a pen randomized controlled trial through the data collection period

	August (6–9 WOT)			September (10–13 WOT)			October (14–16 WOT)		
Item	Min	Mean (%)	Max	Min	Mean (%)	Max	Min	Mean (%)	Max
THI >74									
Farm A	7	13.3 (55.5)	22	0	6.4 (26.8)	19	0	2.4 (10.0)	7
Farm B	9	13.0 (54.0)	18	0	7.1 (29.7)	18	0	2.6 (10.8)	7
THI ≥68									
Farm A	16	22.5 (93.6)	24	6	17.4 (72.5)	24	8	10.8 (45.1)	14
Farm B	14	22.2 (92.4)	24	7	17.5 (72.8)	24	8	10.3 (43.1)	14

For both farms, SCFP diets were formulated to provide a minimum of 19 g/cow per day of SCFP supplement. The hospital pens were fed the SCFP diet. Sick or lame cows were excluded from the study if they spent more than 7 d out of their originally assigned pen. Trained farm staff observed cows for illness at least once daily. We assumed that exposure to SCFP supplement for less than one week by CTRL cows within a hospital or lame pen would not be sufficient to influence the outcomes of interest (Shen et al., 2018).

Table 6. Least squares means (± SE) of dependent variables during the covariate period data (wk 1) of the study in a pen-level randomized controlled trial of supplementation with a *Saccharomyces cerevisiae* fermentation product to lactating diets (SCFP: TMR with SCFP supplementation; or control diet, CTRL: TMR without SCFP supplementation)[1]

Outcome	CTRL	SCFP	*P*-value
Milk yield, kg/cow per d	42.3 ± 2.4	41.9 ± 2.4	0.91
Fat, %	3.60 ± 0.11	3.50 ± 0.11	0.68
Protein, %	2.94 ± 0.04	2.95 ± 0.40	0.82
Lactose, %	4.94 ± 0.02	4.93 ± 0.02	0.95
SNF, %	8.54 ± 0.07	8.54 ± 0.07	0.99
TS, %	12.05 ± 0.17	12.00 ± 0.17	0.84
Linear score	3.51 ± 0.05	3.61 ± 0.05	0.25
Fat yield, kg/cow per d	1.49 ± 0.04	1.45 ± 0.04	0.47
Protein yield, kg/cow per d	1.24 ± 0.06	1.23 ± 0.06	0.93
ECM,[2] kg/cow per d	41.7 ± 1.7	40.9 ± 1.7	0.75
NE_L in milk,[3] Mcal/d	28.9 ± 1.1	28.4 ± 1.1	0.73
DMI,[4] kg/cow per d	24.7 ± 1.3	25.0 ± 1.3	0.85
FE_{MY},[5] kg milk/kg DMI	1.69 ± 0.05	1.64 ± 0.04	0.42
FE_{ECM},[6] kg ECM/kg DMI	1.73 ± 0.04	1.66 ± 0.04	0.41

[1]During this period, both treatment groups were fed the CTRL diet. There were 10 pens enrolled (5 replications per treatment)

[2]ECM = [0.327 × milk yield (kg)] + [12.95 × fat yield (kg)] + [7.65 × protein yield (kg)] (Orth, 1992).

[3]NE_L in milk = [(0.0929 × % milk fat) + (0.0547 × % milk CP) + (0.0395 × % milk lactose)] × milk yield (NRC, 2001).

[4]DMI measured at the pen level.

[5]FE_{MY} = milk yield (kg/cow per d)/DMI (kg/cow per d). Feed efficiency measured at the pen level by using average milk production by pen (arithmetic mean of individual cow milk production for each specific pen) and pen DMI (measured directed at pen level; not individual cow DMI data).

[6]FE_{ECM} = ECM (kg/cow per d)/DMI (kg/cow per d). Feed efficiency measured at the pen level by using average ECM production by pen (arithmetic mean of individual cow ECM production for each specific pen) and pen DMI.

On farm A, test supplements for each diet were delivered to the farms in plain bags of 25 kg labeled as A or B. The CTRL (rice hulls; bag A) or SCFP supplements (bag B; Table 2) were included in a premix containing all other concentrates included in the TMR. The premix was prepared approximately every 4 d on site and stored in different commodity bays labeled as CTRL Premix or SCFP Premix. Because of differences of intake for pens (multiparous vs. primiparous cows), the primiparous cows were provided approximately 20.8 g/cow per day of SCFP (target intake of 24.5 kg DM/cow per day), whereas multiparous were provided approximately 23.4 g/cow per day (target intake of 27.5 kg DM/cow per day).

On farm B, because of on-farm feeding logistics, a control supplement was not added to the CTRL diet. Instead, the lactating diet was mixed for both treatment groups and delivered first to CTRL pens, followed by the addition of the SCFP supplementation. After delivery of CTRL diet to CTRL pens, SCFP supplement was added to the TMR mixer-wagon containing the lactating diet and mixed for 5 min before delivery of the SCFP diet to SCFP pens. Before initiation and during the study, particle size was evaluated in both diets (CTRL and SCFP) on farm B after TMR delivery using the Penn State Particle Separator (Kononoff et al., 2003). The proportion of the TMR retained in the top, middle, lower screen, and pan was similar for both treatment diets, suggesting that the additional mixing time in the SCFP ration did not change the physical characteristics of the diet. The target DMI was 24.5 kg/cow per day.

Formulation changes to the diet were only made based on forage analysis to maintain the nutrient parameters of the TMR. All changes to the lactating base diet were identical for both SCFP and CTRL pens. The inclusion of SCFP was kept the same throughout the study. Diet formulation and nutrient profile fed are presented in Tables 2 and 3, respectively.

Before initiation and during the study period, the distribution of the supplement within the TMR was

evaluated (twice for farm A and 3 times for farm B). The mixer equipment (farm A: Solomix 2 VLH, Triolet; farm B: Tormex 3250 V, Tormex Industrias) and protocol were validated to ensure proper mixing and delivery of supplemented TMR. Results indicated that the mixers were performing as expected with a consistent distribution of the supplement throughout the diets (results not shown).

Weekly samples of the TMR delivered to each pen were collected from the feed bunk using a standardized TMR sampling protocol (Robinson and Meyer, 2010) and frozen at −20°C. Monthly composite samples were created for each pen by thawing and mixing all samples by pen for a given month and subsampling the composited mixture. The composite samples were submitted to a commercial laboratory (Rock River, Torreón, Mexico) for near infrared reflectance spectroscopy. The nutrient composition of the diets based on near infrared reflectance spectroscopy analyses is presented in Table 3.

Data Collection and Processing

Daily milk weights were recorded for each individual animal and daily DMI was recorded by pen. Individual milk yield data and pen DMI were averaged by week of trial (**WOT**) of the data collection period (5–16 WOT). Feed mixing and delivery data were recorded by a commercial feeding software (TMR Tracker, Digi-Star, Topcon Positioning Group). The number of animals per pen was monitored and tracked using the management software linked with TMR Tracker.

Pen refusals were collected, weighed, and recorded daily (feeding software). The target for daily refusals was 3% of previous days feed offered. The pen DMI was calculated daily by the feeding software considering the feed delivered, refused, and the number of animals per pen.

Feed efficiency was calculated at the pen-level using arithmetic average of individual milk production (average of individual cow production for each specific pen) and pen DMI (measured directed at pen level; not individual cow DMI data). Two production measures were used to assess FE: FE_{MY} = milk yield divided by DMI, kg/kg and FE_{ECM} = ECM divided by DMI, kg/kg. Pen ECM production was only calculated for the weeks with individual milk components measurements.

For milk components and somatic cell testing, individual milk samples were collected biweekly (6, 8, 10, 12, 14, and 16 WOT), during the first morning milking shift using in-line milk sampling devices (Ambic). Each sample was divided into 2 aliquots (without preservatives). One aliquot was sent to a commercial laboratory (Laboratorio de Analisis Clinicos Veterinarios Burciaga, Torreón, Mexico) for analysis of milk SCC (Fossomatic 7 DC, FOSS). The other was analyzed on site for milk components (LactoScope FT-B, PerkinElmer). String pen samples were collected weekly and analyzed on site for SCC and milk components using the LactoScope equipment.

Energy corrected milk was calculated as follows: ECM (kg/cow per d) = [0.327 × milk yield (kg)] + [12.95 × fat yield (kg)] + [7.65 × protein yield (kg)] (Orth, 1992). Net energy for lactation (N_{EL} in milk) was calculated as follows: N_{EL} in milk (Mcal/kg) = [(0.0929 × % milk fat) + (0.0547 × % milk CP) + (0.0395 × % milk lactose)] × milk yield (kg)] (NRC, 2001).

All health events, treatments, and moves to the hospital pen were recorded in a standardized format in the herd management software (Afifarm, Afimilk, Kibbutz Afikim; and Dairy Comp 305 Valley Agricultural Softwareon Farms A and B, respectively) and transferred daily to the Dairy Health and Management Services, LLC (Lowville, New York) data warehouse. Health records, including event date, presumptive diagnosis, treatments, and culling were recorded in each herd management software. Health events analyzed were clinical mastitis (visibly abnormal milk clots, flakes, or watery or bloody with or without swelling of the udder confirmed by fore stripping at each milking), pneumonia (diagnosed by the herd veterinarians), and culling (sold and dead). Research support staff observed cows, monitored general status of the facilities, research protocol management, and data collection several times a week before and during the study. Body condition score (1–5 scale with increments of 0.25) was assessed and recorded at the beginning of the study, in the middle, and at the end of the data collection period by the same trained researcher (Edmonson et al., 1989; Ferguson et al., 1994).

Environmental sensors (Kestrel 5400 Heat Stress Tracker, Kestrel Instruments) were placed 2 m above the headlocks in the middle of each pen in both farms during the data collection period. Environment data were collected every 10 min. The THI was calculated based on the NRC formula (NRC, 1971) as follows:

$$THI = (1.8 \times AT_{avg} + 32) - [(0.55 - 0.0055 \times RH_{avg}) \times (1.8 \times AT_{avg} - 26)],$$

where AT_{avg} is the daily average ambient temperature (°C) and RH_{avg} is the daily average relative humidity (%). The THI data were averaged by the hour of each day, then by day, and then by WOT. Descriptive statistics (mean ± SD, maximum and minimum) for THI values are presented by WOT and by month of study (August, September, and October). Exposure to HS was assessed based on the following THI categories: <68

THI: outside the thermal danger zone for cows; 68 to 74 THI: mild signs of HS; ≥75 THI: drastic decreases in production performance (De Rensis et al., 2015). To evaluate the number of hours per day that cows were exposed to a specific THI range (≥68 and >74) we classified each data point (every 10 min) into being above or below the threshold. A specific hour of a specific day was considered above the threshold if 80% of the included data points within that hour were above the respective threshold. For each date (day) we counted the total number of hours above the respective threshold. Within each month (August, September, and October) the minimum, mean, and maximum number of hours exposed to HS are presented. This description is stratified by farm.

Statistical Analysis

All the analyses were conducted using SAS 9.4 (version 9.4, SAS Institute Inc.) following the recommendations for pen-level studies (St-Pierre, 2007). In all models, pen was the experimental unit (pen nested within farm and treatment included as a random effect). The correct denominator degrees of freedom (**DDF**; approximately 8) for the experimental units in the study was obtained by including the random effect of pen nested within farm × treatment. The Kenward-Roger degrees of freedom approximation option (DDFM = KENWARDROGER) was included in all models. The observational unit (pen vs. the individual animal) varied with the outcome assessed. For milk yield, milk components, linear score, health events, and BCS models the observational unit was the cow. For DMI and FE models, the observational unit was the pen. Time was considered WOT for daily milk weights, biweekly milk components, and biweekly linear score models; for the BCS model time was considered as first, second, or third assessment. The effect of treatment (CTRL or SCFP), WOT, and treatment by WOT interaction, parity (primiparous vs. multiparous cows), treatment by parity and the triple-way interaction between treatment, WOT, and parity were offered as fixed effects. When a triple-way interaction of treatment by WOT and parity was detected, both 2-way interactions involving treatment effect (treatment by parity and treatment by WOT) were kept in the model. When an interaction of treatment by parity, treatment by parity and WOT, or WOT was detected, data were stratified and presented by parity, respectively. Baseline values (WOT 1, covariate period) for the outcome of interest (milk yield, milk components, DMI, and FE) were offered as a covariate in their respective statistical models. Continuous outcomes repeated over time [daily milk yield, biweekly milk components, linear score (**LS**), and BCS; cow as the observational unit) were analyzed by ANOVA accounting for repeated measures (MIXED procedure of SAS]. Normality of the residuals fit was assessed with the Shapiro-Wilk test; transformations were not necessary. For daily milk yield, milk components, and LS, the model build is represented by the following linear mixed model equation:

$$Y_{ijkt} = \mu + \beta_i + \gamma_j + \theta_k + \delta_t + (\beta\gamma)_{ij} + (\beta\theta)_{ik} + (\beta\gamma\theta)_{ijk} + F_{pli} + \alpha_{m(ip)} + \varepsilon_{ijkm},$$

where Y_{ijkt} = response to treatment i (CTRL or SCFP) at the kth WOT for m cow with t covariate value; μ = overall mean; β_i = fixed effect of treatment; γ_j = fixed effect of parity; θ_k = fixed effect WOT; δ_t = fixed effect of covariate value for the respective outcome; $(\beta\gamma)_{ij}$ = effect of treatment by parity interaction; $(\beta\theta)_{ik}$ = effect of treatment by WOT interaction; $(\beta\gamma\theta)_{ijk}$ = effect of treatment by parity by WOT interaction; F_{pli} = random effect of pen p nested within farm l × treatment i; $\alpha_{m(ip)}$ = repeated measures, m cow nested within the treatment i and pen p effect; ε_{ijkm} = residual error within cow m, on treatment i, with parity j at WOT k. The main effect of treatment had approximately 8 DDF for all models of milk assessed outcomes. For BCS the model build is represented by the following linear mixed model equation:

$$Y_{ijk} = \mu + \beta_i + \gamma_j + \theta_k + (\beta\gamma)_{ij} + (\beta\theta)_{ik} + (\beta\gamma\theta)_{ijk} + F_{pli} + \alpha_{m(pi)} + \varepsilon_{ijkm},$$

where Y_{ijk} = response to treatment i (CTRL or SCFP) at the kth time of assessment for m cow with j parity; μ = overall mean; β_i = fixed effect of treatment; γ_j = fixed effect of parity; θ_k = fixed effect time of assessment; $(\beta\gamma)_{ij}$ = effect of treatment by parity interaction; $(\beta\theta)_{ik}$ = effect of treatment by time of assessment interaction; $(\beta\gamma\theta)_{ijk}$ = effect of treatment by parity by time of assessment interaction; F_{pli} = random effect of pen p nested within farm l × treatment i; $\alpha_{m(pi)}$ = repeated measures, m cow nested within the treatment i and pen p effect; ε_{ijkm} = residual error within cow m, on treatment i, with parity j at k time of assessment. In models where pen was the observational unit (DMI and FE; absence of cow-level intake data), parity effect and its interaction with treatment were not offered as fixed effect because on farm B primiparous and multiparous cows were commingled. Dry matter intake was measured at the pen-level and DMI and FE were analyzed by ANOVA with repeated measures (MIXED procedure of SAS). Pen within treatment was the subject of repeated measures over time. For DMI and FE the model build is represented by the following linear mixed model equation:

$$Y_{ikt} = \mu + \beta_i + \theta_k + \delta_t + (\beta\theta)_{ik} + F_p + \alpha_{p(il)} + \varepsilon_{iklp},$$

where Y_{ikt} = response to treatment i (CTRL or SCFP) at the kth WOT for p pen with t baseline value; μ = overall mean; β_i = fixed effect of treatment; θ_k = fixed effect of WOT; δ_t = fixed effect of covariate value for the respective outcome; $(\beta\theta)_{ik}$ = effect of treatment by WOT interaction; F_l = random effect of farm l; $\alpha_{p(il)}$ = repeated measures of p pen nested within the treatment i and farm l effect; ε_{iklp} = residual error within pen p and treatment i, at week k. The main effect of treatment had approximately 8 DDF for all models of FE and DMI outcomes. A windstorm was reported on the day farm A tested milk components and SCC for the 16 WOT. Due to the strong environmental changes, data were analyzed with and without this data point. We removed WOT 16 data from these analyses because it drastically changed the results observed until WOT 14.

Dichotomous outcomes (clinical mastitis, pneumonia, and culling) were analyzed by logistic regression (GLIMMIX procedure of SAS). The effect of treatment (CTRL or SCFP), parity (primiparous vs. multiparous cows), and treatment by parity interaction were offered as fixed effects to all models. Logistic regression models are represented by the following equation:

$$y_{ij} \mid \pi_{ij} \sim \text{Binary}\ (1,\ \pi_{ij})$$

$$\pi_{ij} = \text{P}\ (\pi_{ij} = 1 \mid \beta_i,\ \gamma_j,\ F_{pli})$$

$$\text{logit}\ \{\pi_{ij}\} = \beta_i + \gamma_j + (\beta\gamma)_{ij} + F_{pli},$$

where π_{ijl} represents the probability (P) of a cow at j parity on i treatment having the outcome of interest, i = treatment (CTRL or SCFP), j = parity group, β_i = fixed effect of treatment, γ_j = fixed effect of parity, $(\beta\gamma)_{ij}$ = effect of treatment by parity interaction, F_{pli} = random effect of pen p nested within farm $l \times$ treatment i.

The main effect of treatment had 8 DDF for all models of health outcomes. Logistic regression model results are presented as odds ratio with 95% confidence intervals.

The reduced model for each outcome of interest was selected by backward elimination of covariates with $P > 0.10$. Covariate structure for repeated measures was selected based on the lowest Aikaike Information Criterion. In the presence of significant interaction terms (e.g., treatment by WOT), LSM and statistical significance of pair-wise comparisons of interest were obtained by applying the PDMIX800 macro (Saxton, 1998) with the default adjustment option. Data were stratified by parity group in the presence of an interaction of treatment by parity. Explanatory variables and their interactions were considered significant if $P \leq 0.05$, and $0.05 < P \leq 0.10$ was considered a tendency.

RESULTS

A total of 10 lactating pens (experimental units; 5 CTRL pens; 5 SCFP pens) from 2 commercial dairy farms were included in this randomized controlled trial (Table 1). A total of 1,843 cows (915 primiparous and 928 multiparous cows) were enrolled. From the number of cows enrolled, 78 cows did not complete the experiment because they were moved to the hospital pens and stayed longer than 7 d (60 cows from farm A; 18 from farm B; 43 from CTRL; 35 from SCFP groups), and 120 cows were culled (32 cows from farm A; 88 from farm B; 64 from CTRL; 56 from SCFP groups). From those cows that did not complete the experiment, 27 of them were culled (26 sold and 1 died) before the data collection period (19 cows from farm A; 8 cows from farm B; 16 from CTRL; 11 from SCFP groups). All 19 cows sold in farm A before the data collection period (23% of the total cows culled during the experiment) were removed from the experiment because they tested positive for tuberculosis as part of an eradication program, but they were otherwise healthy, productive cows.

A total of 1,645 cows (839 primiparous and 806 multiparous cows) completed the WOT 16 of the study. At enrollment, lactation number, DMI, and milk yield were not different between CTRL and SCFP pens (Table 4). Milk yield, milk components, and FE during the covariate period are presented by treatment group in Table 6.

Description of the THI

Cows were exposed to a THI ≥68 during the months of August, September, and October, for 100, 87, and 21% of the days each month, respectively. During the study period, the daily maximum THI was always above THI 74 except for 2 d, one each in September and October (Figure 1). The average number of hours per day spent in the danger zone (THI ≥68) was greater than 22 h in August with more than 13 h per day presenting a THI >74 (Table 5). In October, we observed fewer hours per day of HS conditions (10 h 30 min per day on average; Table 5).

Effect of Treatment on Milk Yield and Components

An interaction of treatment by WOT within parity group was detected for milk yield, ECM, and NE_L (Table 7). Results from these models were stratified by

Table 7. Least squares means (± SE) of dependent variables during the data collection period (WOT; 5–16 wk of the study) in a pen-level randomized controlled trial of supplementation with a *Saccharomyces cerevisiae* fermentation product to lactating diets (SCFP: TMR with SCFP supplementation; or control diet, CTRL: TMR without SCFP supplementation)[1]

	Treatment (Trt)		*P*-value						
Outcome	CTRL	SCFP	Trt[2]	Baseline[3]	WOT[4]	Parity[5]	Trt × WOT	Trt × parity	Trt × WOT × parity[4]
Milk yield, kg/cow per d	38.84 ± 0.63	39.24 ± 0.63	0.66	<0.0001	<0.0001	0.09	<0.0001	0.44	<0.0001
Fat, %	3.99 ± 0.08	4.06 ± 0.08	0.56	<0.0001	<0.0001	—	—	—	—
Protein, %	3.24 ± 0.02	3.25 ± 0.02	0.81	<0.0001	<0.0001	0.03	—	—	—
Lactose, %	4.85 ± 0.01	4.86 ± 0.01	0.67	<0.0001	<0.0001	<0.0001	—	—	—
SNF, %	8.81 ± 0.02	8.83 ± 0.02	0.46	<0.0001	<0.0001	0.006	—	—	—
TS, %	12.82 ± 0.11	12.91 ± 0.11	0.63	<0.0001	<0.0001	—	—	—	—
Linear score	2.54 ± 0.11	2.49 ± 0.11	0.78	<0.0001	<0.0001	<0.0001	—	—	—
Fat yield, kg/cow per d	1.56 ± 0.03	1.59 ± 0.03	0.51	<0.0001	0.02	—	—	—	—
Protein yield, kg/cow per d	1.25 ± 0.02	1.26 ± 0.02	0.72	<0.0001	<0.0001	0.07	—	—	—
ECM,[6] kg/cow per d	42.38 ± 0.70	43.06 ± 0.70	0.51	<0.0001	<0.0001	0.08	0.77	0.49	<0.001
NE_L in milk,[7] Mcal/d	28.72 ± 0.48	29.22 ± 0.48	0.49	<0.0001	<0.0001	0.08	0.68	0.38	0.001
DMI,[8] kg/cow per d	26.0 ± 0.21	25.2 ± 0.21	0.01	<0.0001	0.001	NA	—	NA	NA
FE_{MY},[9] kg milk/kg DMI	1.53 ± 0.05	1.59 ± 0.05	0.003	0.003	<0.0001	NA	—	NA	NA
FE_{ECM},[10] kg ECM/kg DMI	1.68 ± 0.02	1.73 ± 0.02	0.02	—	<0.0001	NA	—	NA	NA

[1]There were 10 pens enrolled (5 replications per treatment). The dash indicates fixed effect included in the full model but removed ($P > 0.1$) during the backward elimination process. NA = not applicable; parity effect was not offered to model with pen as observational unit because of the parity distribution at farm B.

[2]Effect of supplementation of lactating TMR with a *Saccharomyces cerevisiae* fermentation product. The denominator degrees of freedom (DDF) for the effect of treatment accounted for the clustering of animals. For all models presented, DDF varied between 6.0 to 8.19.

[3]For each model, the baseline covariate for the respective outcomes assessed during baseline week of the study (wk 1) was offered to the model.

[4]WOT = week of trial. Data collection period: wk 5 to wk 16; for milk components and SCC results, data were collected biweekly. Week 16 of the trial was removed from the analysis of because of environmental issues on the day of testing in farm A (strong windstorm) that dramatically affected the results of that test.

[5]Effect of parity (primiparous vs. multiparous cows).

[6]ECM = [0.327 × milk yield (kg)] + [12.95 × fat yield (kg)] + [7.65 × protein yield (kg)] (Orth, 1992).

[7]NE_L in milk = [(0.0929 × % milk fat) + (0.0547 × % milk CP) + (0.0395 × % milk lactose)] × milk yield (NRC, 2001).

[8]DMI measured at the pen level.

[9]FE_{MY} = milk yield (kg/cow per d)/DMI (kg/cow per d). Feed efficiency measured at the pen level by using average milk production by pen (arithmetic mean of individual cow milk production for each specific pen) and pen DMI (measured directed at pen level; not individual cow DMI data).

[10]FE_{ECM} = ECM (kg/cow per d)/DMI (kg/cow per d). Feed efficiency measured at the pen level by using average ECM production by pen (arithmetic mean of individual cow ECM production for each specific pen) and pen DMI.

parity group (Figures 2, 3, and 4). In primiparous cows, milk yield, ECM, and NE_L were not different between treatment groups at any WOT (Figures 2a, 3a, and 4a, respectively). In multiparous cows, the SCFP group yield more milk on WOT 5 (SCFP: 44.6 ± 0.28 kg/cow per day; CTRL: 43.4 ± 0.27), WOT 7 (SCFP: 44.0 ± 0.28; CTRL: 43.0 ± 0.28), WOT 10 (SCFP: 40.4 ± 0.28; CTRL: 39.4 ± 0.28), WOT 11 (SCFP: 41.2 ± 0.28; CTRL: 40.2 ± 0.28), WOT 14 (SCFP: 41.5 ± 0.28; CTRL: 40.4 ± 0.29), and WOT 15 (SCFP: 41.2 ± 0.29; CTRL: 40.4 ± 0.29, $P < 0.05$) compared with multiparous cows of the CTRL group (average difference of 0.8 kg/cow per day; Figure 2). Multiparous cows of SCFP group had a constant numerically greater ECM and NE_L compared with cows of CTRL group (average +1.25 kg/cow per day, $P = 0.14$; +0.89 Mcal/d, $P = 0.11$, respectively; Figures 3 and 4) during all weeks of the study. Differences between treatment groups were not detected for milk components yield or percentage ($P > 0.40$; Table 7).

Effect of Treatment on DMI, Feed Efficiency, and BCS

Dry matter intake and FE were assessed at the pen level. Treatment by WOT interaction was not significant for DMI ($P = 0.99$; Table 7). Pens of cows fed SCFP diet had lower DMI than pens of cows fed the CTRL diet (25.2 ± 0.2 kg/cow per day vs. 26.0 ± 0.2 kg/cow per day, $P = 0.01$).

The interaction of treatment and WOT was not significant for either measure of FE ($P = 0.99$ and $P = 0.56$, respectively). Feed efficiency was greater for pens fed SCFP compared with pens fed CTRL (FE_{MY} 1.59 ± 0.05 vs. 1.53 ± 0.05, $P = 0.003$, respectively; FE_{ECM} 1.73 ± 0.02 vs. 1.68 ± 0.02, $P = 0.02$, respectively).

The mean DIM (± SD) for BCS at first, second, and third assessments were 153 ± 54, 189 ± 54, 245 ± 54 DIM, respectively. An interaction of treatment by time of assessment within parity group was detected ($P < 0.0001$). At the first and second assessments, BCS was

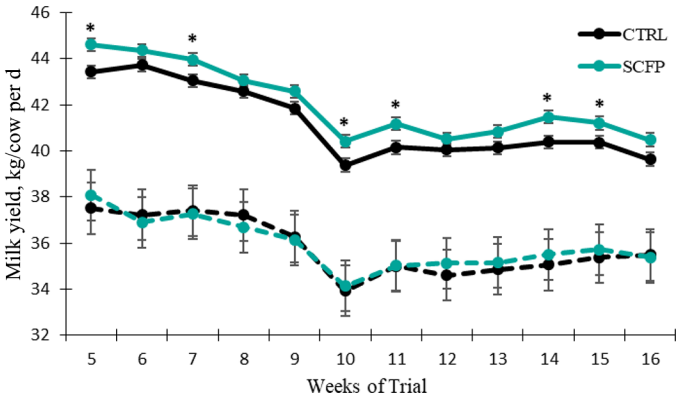


Figure 2. Least squares means (± SE) of milk yield (kg/cow per day) during the data collection period of the study (weeks of trial, WOT) for primiparous (dashed lines) and multiparous (solid lines) cows in a pen-level randomized controlled trial of supplementation with a *Saccharomyces cerevisiae* fermentation product to lactating diets (SCFP: TMR with SCFP supplementation; or control diet, CTRL: TMR without SCFP supplementation). Individual milk yield data were collected daily and averaged by WOT. There were interactions of treatment with parity and WOT. For multiparous cows, within WOT, * denotes an effect of SCFP ($P < 0.05$). There were 10 pens enrolled (5 replications per treatment).

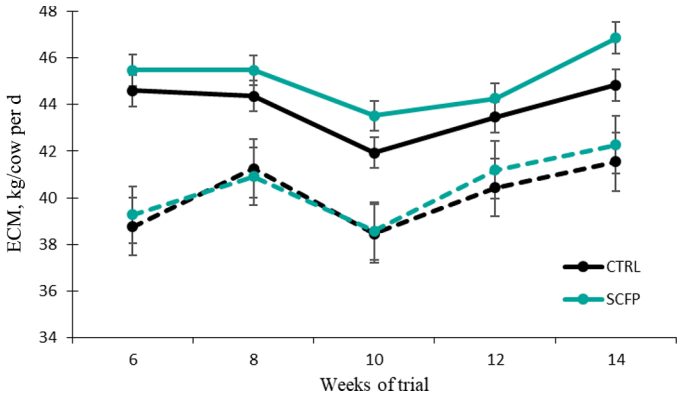


Figure 3. Least squares means (± SE) of ECM (kg/cow per day) during the data collection period of the study (weeks of trial, WOT; biweekly assessment) for primiparous (dashed lines) and multiparous (solid lines) cows in a pen-level randomized controlled trial of supplementation with a *Saccharomyces cerevisiae* fermentation product to lactating diets (SCFP: TMR with *Saccharomyces cerevisiae* fermentation product supplementation; or control diet, CTRL: TMR without supplementation). There were interactions of treatment with parity and WOT. There were 10 pens enrolled (5 replications per treatment). Week 16 of the trial was removed from the analysis because of environmental issues on the day of testing in farm A (strong windstorm) that dramatically affected the results of the test.

not different between treatment groups ($P > 0.69$). Differences were observed at the third BCS assessment in both parity groups (primiparous cows: CTRL 3.23 ± 0.02 vs. SCFP 3.33 ± 0.02, $P = 0.006$; multiparous cows: CTRL 3.04 ± 0.02 vs. SCFP 3.11 ± 0.02, $P = 0.05$; Figure 5).

Mastitis, Pneumonia, and Culling

Multiparous cows had a greater incidence of clinical mastitis [multiparous cows = 7.0 ± 2.3%; primiparous cows = 2.3 ± 1.1%; adjusted odds ratio (**Adj. OR**) = 3.18 (95% CI = 1.22–8.33); $P = 0.018$] and a greater proportion left the herd [multiparous cows = 4.5 ± 1.8%; primiparous cows = 2.2 ± 1.0%; Adj. OR = 2.12 (95% CI: 1.30–3.44); $P = 0.003$] compared with primiparous cows. Clinical mastitis [CTRL: 4.5 ± 2.0% vs. SCFP: 3.6 ± 1.7%; Adj. OR = 1.28 (0.29–5.80); $P = 0.71$], pneumonia [CTRL 1.6 ± 1.2%; SCFP: 1.7 ± 1.3%; Adj. OR = 0.90 (0.08–10.71); $P = 0.92$], and culling events [CTRL 2.9 ± 1.7%; SCFP: 3.5 ± 2.0%; Adj. OR = 0.81 (0.12–5.60); $P = 0.81$] were not different between the treatment groups (Table 8).

DISCUSSION

To our knowledge, this was the first randomized controlled trial performed on a large scale in commercial dairy farms including lactating cows exposed to high temperature and humidity conditions. This study was performed in the northern part of Mexico, where cows experience severe environmental conditions during summer months. As hypothesized, feeding a SCFP improved FE in lactating Holstein cows exposed to environment conditions outside the thermoneutral zone. Because of the study design, it was not possible to as-

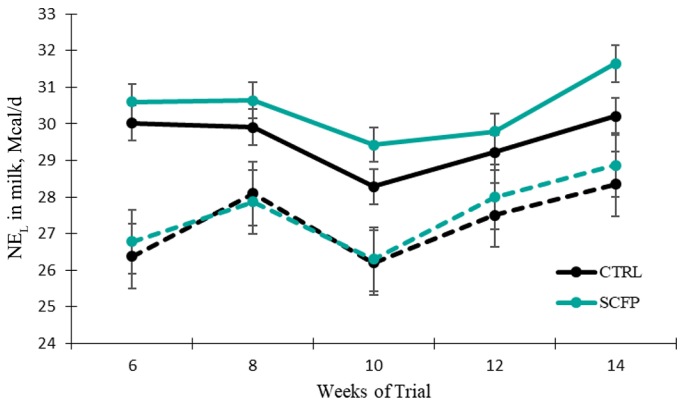


Figure 4. Least squares means (± SE) of NE_L in milk (Mcal/d) during the data collection period of the study (weeks of trial, WOT; biweekly assessment) for primiparous (dashed lines) and multiparous (solid lines) cows in a pen-level randomized controlled trial of supplementation with a *Saccharomyces cerevisiae* fermentation product to lactating diets (SCFP: TMR with *Saccharomyces cerevisiae* fermentation product supplementation; or control diet, CTRL: TMR without supplementation). There were interactions of treatment with parity and WOT. There were 10 pens enrolled (5 replications per treatment). Week 16 of the trial was removed from the analysis because of environmental issues on the day of testing in farm A (strong windstorm) that dramatically affected the results of the test.

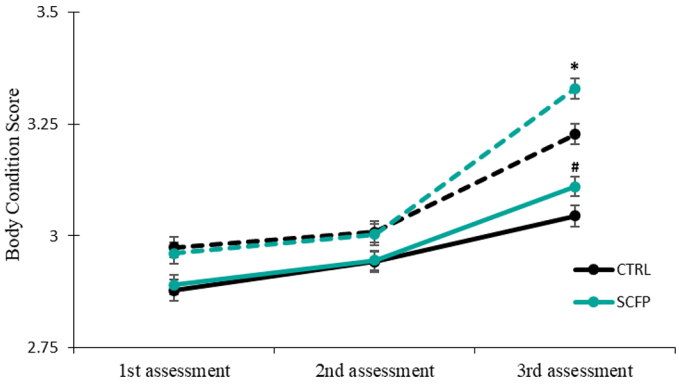


Figure 5. Least squares means (± SE) of BCS (0–5 scale with 0.25-point increment) in a pen-level randomized controlled trial of cows a TMR supplemented with *Saccharomyces cerevisiae* fermentation product (SCFP) or without supplementation (CTRL) at first (153 DIM ± 54 SD), second (189 DIM ± 54 SD), and third (245 DIM ± 54 SD) BCS assessments. The experimental unit was the pen (5 replications per treatment). An interaction of treatment by parity and time of assessment was detected ($P < 0.0001$); results were stratified by parity group (primiparous: dashed lines; multiparous cows: solid lines). Within parity group and at each assessment time, * denotes the effect of *Saccharomyces cerevisiae* fermentation product supplementation at $P = 0.006$; # denotes a $P = 0.05$.

sess this outcome based on parity. Although milk yield differences were not observed in primiparous cows, the overall greater FE can be explained through lower DMI without a reduction in milk production. Statistical differences in milk components and milk component yields, LS, mastitis and pneumonia, and culling events were not detected between treatment groups.

Extensive work has been published regarding the effect of SCFP on milk production, DMI, physiological parameters, and inflammation biomarkers when fed at different rates and stages of lactation (Poppy et al., 2012; Olagaray et al., 2019). Feed efficiency, a more comprehensive indicator of performance, is a function of milk production and feed intake, and ultimately the driver of a dairy farm's profit (e.g., income over feed cost). Thus, we consider it important to discuss the effects of supplementation with SCFP both on milk production and DMI.

In a meta-analysis of feeding SCFP to lactating cows (Poppy et al., 2012), the association between DMI and supplementation with SCFP was different in early lactation versus other stages of lactation. In early lactation studies, supplementation was associated with an increase in DMI (+0.62 kg/d; 95% CI: 0.21–1.02), whereas in later lactation studies DMI decreased in cows supplemented with SCFP [−0.78 kg/d; (−1.36 to −0.21)]. Our study included cows from an average of 100 to 245 DIM and we observed that DMI decreased for pens supplemented with SCFP compared with the CTRL group (average difference of 0.8 kg/d). A DMI decrease after the fresh period represents greater FE when milk yield is maintained or increased (Poppy et al., 2012), as observed in the current study. Results from Acharya et al. (2017) are also in agreement; these

Table 8. Summary of multivariable logistic regression models of health performance of Holstein cows in a pen-level randomized controlled trial of supplementation with a *Saccharomyces cerevisiae* fermentation product to lactating diets (SCFP: TMR with SCFP supplementation; or control diet, CTRL: TMR without SCFP supplementation)[1]

	Treatment			*P*-value	
Outcome, %	CTRL	SCFP	Adj. OR[2] (95% CI)	Treatment[3]	Parity[4]
Clinical mastitis[5]	4.5 ± 2.0	3.6 ± 1.7	0.78 (0.17–3.45)	0.71	0.02
Pneumonia[6]	1.6 ± 1.2	1.7 ± 1.3	1.11 (0.09–12.5)	0.92	0.92
Sold	2.7 ± 1.6	3.4 ± 1.9	1.27 (0.20–8.33)	0.78	0.003
Died	0.4 ± 0.3	0.1 ± 0.1	0.29 (0.01–6.67)	0.38	0.72
Left the herd[7]	2.9 ± 1.7	3.5 ± 2.0	1.23 (0.18–8.33)	0.81	0.003

[1]There were 10 pens enrolled (5 replications per treatment). Results are presented as LSM ± SE. Treatment by parity interaction was not detected in any of the models ($P > 0.1$).

[2]CTRL group is the referent (ref.).

[3]Effect of supplementation of lactating TMR with a *Saccharomyces cerevisiae* fermentation product. For all models presented, the denominator degrees of freedom for the effect of treatment was 8.

[4]Effect of parity: primiparous (ref.) vs. multiparous cows. Clinical mastitis: multiparous cows 7.0 ± 2.3%; primiparous cows 2.3 ± 1.1%; Adjusted odds ratio (Adj. OR) = 3.18 (95% CI = 1.22–8.33); Sold: multiparous cows 4.4 ± 1.8%; primiparous cows 2.1 ± 0.9%; Adj. OR = 2.15 (1.30–3.55); Left the herd: multiparous cows 4.5 ± 1.8%; primiparous cows 2.2 ± 1.0%; Adj. OR = 2.12 (1.30–3.44).

[5]Visibly abnormal milk clots, flakes, or watery or bloody with or without swelling of the udder confirmed by fore-stripping at each milking.

[6]Diagnosed by the herd veterinarians.

[7]Sold and dead.

authors described either numerically (1.34 vs. 1.30) or significantly (1.41 vs. 1.30) greater FE (Milk/DMI) when feeding SCFP to lactating cows compared with the control. The reduced DMI in the SCFP group could be explained by the hepatic oxidation theory and the hypophagic effect of propionate (Allen et al., 2009). Miller-Webster et al. (2002) have demonstrated that SCFP increased ruminal production of propionate. The increased propionate entry into the liver provides the Krebs cycle intermediates that allow oxidation of acetyl-CoA. Oxidizing the pool of acetyl-CoA rather than exporting it increases ATP production and likely causes satiety despite the use of propionate for glucose synthesis (Allen et al., 2009). We did not collect individual cow data that would support this theory.

A meta-analysis (Poppy et al., 2012) including 21 peer-reviewed studies on the effect of SCFP on milk production demonstrated an overall increased milk yield when adding SCFP to the TMR of lactating cows [+1.18 kg/d (95% CI: 0.55–1.81)] with a significant increase of fat and protein yield (+0.06 kg/d and +0.03 kg/d, respectively). In a study with 80 mid-lactation cows, Acharya et al. (2017) compared the effect of the same product fed in our study versus a control TMR without supplementation and reported an increased milk yield (36.8 vs. 33.3 kg/d ± 2.38). In terms of milk components, the meta-analysis demonstrated an increase in fat and protein yield (+0.06 kg/d and +0.03 kg/d, respectively; Poppy et al., 2012). Acharya et al. (2017) reported a reduction in fat concentration (3.85 vs. 4.17% ± 0.11) and an increase in lactose yield (1.82 vs. 1.64 kg/d ± 0.14) in the supplemented versus control groups. We have not observed milk component differences between treatment groups. The fact that milk components were measured biweekly with a single milking sample instead of daily (as in the referenced studies) likely made it more difficult to detect differences between groups. The greater milk yield with no differences detected in milk components still resulted in a greater ECM, which was consistently greater in the SCFP than in the control cows throughout the data collection period.

An interaction of treatment by parity was detected in some of the outcomes studied (e.g., milk yield and ECM). The distinct nutrient and energy requirements between primiparous and multiparous cows are well described (NRC, 2001). Considering the environmental conditions during our study, we consider tolerance to HS across parities (primiparous being more resistant to heat stress than multiparous cows) an important topic to discuss. For example, Bernabucci et al. (2015) reported a maximum yield loss of 0.91, 1.16, and 1.27 kg/d for first, second, and third lactation cows, respectively, when exposed to HS conditions. These differences may be explained by a lower generation of metabolic heat, a greater proportion of surface area to dissipate heat, and lower milk yield in primiparous vs. multiparous cows (Bernabucci et al., 2015). Thus, we were not surprised by the different magnitude of response to SCFP observed between primiparous and multiparous cows. Although milk production was not different between treatment groups of primiparous cows, FE was improved in both parity groups. This improvement was likely driven by DMI, but unfortunately, due to the management of the farms, DMI was not possible to assess by parity.

Primiparous and multiparous cows fed SCFP had greater BCS at the end of the study period than the respective controls; in multiparous cows, SCFP effect was accompanied by a greater milk yield response compared with counterparts fed the CTRL ration. Other studies have evaluated BCS dynamics when feeding a SCFP to transition and lactating dairy cows. Zhu et al. (2016) (n = 81 mid-lactation multiparous cows) reported that cows fed SCFP increased BCS throughout the study, while control cows decreased. However, other authors did not detect BCS differences between study groups (Acharya et al., 2017; Olagaray et al., 2019; Shi et al., 2019). The discrepancy of SCFP effect on BCS among the studies could be due to the differences in energy partitioning between milk production and BCS. For the studies where SCFP supported more milk (numerical or significant), BCS was similar between the treatments (Acharya et al., 2017; Olagaray et al., 2019; Shi et al., 2019), whereas when no milk response was observed, the increase in BCS was more apparent (Zhu et al., 2016). In addition, those studies showing no difference in BCS were performed for 6 to 8 wk of lactation. In our study, BCS did not differ until the final measurement at 15 wk. Thus, it may take longer than 8 wk on treatment for lactating cows to detect a difference in BCS. In our study, differences relative to BCS among treatment groups were observed for both parity groups, more noticeably in the primiparous group. We speculate that although multiparous cows used the additional available energy for milk production, primiparous cows did so toward growth. We did not collect data to support this hypothesis. Greater BCS in the treatment group cows could be explained by greater metabolic efficiency and greater net energy balance (Zhu et al., 2016), where a greater proportion of energy was obtained from the diet when compared with control cows. More specifically, our hypothesis is that cows fed SCFP increased energy supply (e.g., VFA; Miller-Webster et al., 2002), which can be used to support milk production, BCS in late lactation (while milk production is diminishing), or both. As primiparous cows are still growing, more energy could be used to support BCS

with little effect on milk yield. For multiparous cows, an increased energy supply was probably allocated to support milk yield with a moderate increase in BCS. Our overall results may indicate that supplementation with SCFP can improve nutrient utilization and support the replenishment of body reserves at the later stages of lactation. The greater BCS in late lactation, close to dry-off, could be advantageous for subsequent lactation productivity.

Challenges and Study Limitations

As expected, the magnitude of exposure to HS conditions decreased throughout the study. The range of time exposed to HS conditions varied widely (from a maximum of 24 h/d outside the thermoneutral zone (THI $\geq$68) in August and September to a minimum of 8 h/d in October). In controlled conditions, Wheelock et al., (2010) studied the effects of exposure to HS on energetic metabolism. The thermoneutral group was composed of cows (n = 10) always exposed to a <67 THI. Their HS group included cows that were always exposed to a THI >70, with 12 h/d exposed to THI >75 (Wheelock et al., 2010). Although we could not control the environmental conditions, the THI variation experienced by cows in our study reflects the conditions experienced on commercial dairy farms. On these commercial farms, some management practices are applied to mitigate HS (e.g., sprinklers and fans) and those were maintained during the study. We believe that these practices reduced the magnitude of the adverse effects of HS throughout the study, but these represent common HS mitigation strategies, increasing the external validity of the results obtained. As these practices were equally applied to both treatment groups, they should not have interfered with the study conclusions.

Because of the study design, we cannot conclude on the level of HS experience by individual animals, but rather that cows were exposed to THI levels outside of what has been described as their thermoneutral zone.

On farm B, due to practical reasons, it was not possible to prepare 2 TMR diets (SCFP and CTRL) individually. To overcome this challenge, on this farm the SCFP supplement was added to the prepared CTRL TMR after the CTRL diet was delivered to the CTRL pens. Additional mixing time in preparation of SCFP TMR may have contributed to additional differences among the SCFP and CTRL rations that we were not able to control. However, particle size separation evaluations performed throughout the study were not different between the 2 rations (before and after the addition of SCFP supplement). These results, along with the results for mixer validation (twice for farm A and 3 times for farm B), Microtracer F evaluation (twice a week in both farms), and weekly TMR sampling for each pen (both farms), suggest that the rations were similar between treatment groups except for the addition of SCFP and highlight our efforts to ensure the treatments were comparable and applied correctly.

While on farm A parity segregation was possible given the infrastructures and farm size, on farm B primiparous and multiparous cows were commingling in the same pens. This practical limitation removed the ability to evaluate the effect of pen-level outcomes (DMI, FE) by parity group. Future pen-level studies should evaluate milk yield, DMI, and FE secondary to feeding SCFP considering complete parity segregation to better understand the potential different magnitude of responses among cow populations.

CONCLUSIONS

In this pen-level randomized controlled trial, lactating cows supplemented with a SCFP during a period of greater risk for HS had greater milk production (multiparous cows), improved FE and increased BCS in the late stage of lactation than cows exposed to the same environmental conditions not fed SCFP. Feeding SCFP to lactating cows demonstrated to be an effective practice for the mitigation of production losses during extreme environmental temperatures on commercial dairy farms.

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REFERENCES

Acharya, S., J. P. Pretz, I. Yoon, M. F. Scott, and D. P. Casper. 2017. Effects of *Saccharomyces cerevisiae* fermentation products on the lactational performance of mid-lactation dairy cows. Transl. Anim. Sci. 1:221–228. https://doi.org/10.2527/tas2017.0028.

Al-Qaisi, M., E. A. Horst, E. J. Mayorga, B. M. Goetz, M. A. Abeyta, I. Yoon, L. L. Timms, J. A. Appuhamy, and L. H. Baumgard. 2020a. Effects of a *Saccharomyces cerevisiae* fermentation product on heat-stressed dairy cows. J. Dairy Sci. 103:9634–9645. https://doi.org/10.3168/jds.2020-18721.

Al-Qaisi, M., E. J. Mayorga, E. A. Horst, S. K. Kvidera, C. S. McCarthy, M. A. Abeyta, B. M. Goetz, H. A. Ramirez-Ramirez, L. L. Timms, and L. H. Baumgard. 2020b. Validating a heat stress model: The effects of an electric heat blanket and nutritional plane on lactating dairy cows. J. Dairy Sci. 103:5550–5560. https://doi.org/10.3168/jds.2019-17543.

Allen, M. S., B. J. Bradford, and M. Oba. 2009. Board-invited review: The hepatic oxidation theory of the control of feed intake and its

application to ruminants. J. Anim. Sci. 87:3317–3334. https://doi.org/10.2527/jas.2009-1779.
Becker, C. A., R. J. Collier, and A. E. Stone. 2020. Invited review: Physiological and behavioral effects of heat stress in dairy cows. J. Dairy Sci. 103:6751–6770. https://doi.org/10.3168/jds.2019-17929.
Bello, N. M., M. Kramer, R. J. Tempelman, W. W. Stroup, N. R. St-Pierre, B. A. Craig, L. J. Young, and E. E. Gbur. 2016. Short communication: On recognizing the proper experimental unit in animal studies in the dairy sciences. J. Dairy Sci. 99:8871–8879. https://doi.org/10.3168/jds.2016-11516.
Bernabucci, U., L. Basiricò, P. Morera, D. Dipasquale, A. Vitali, F. Piccioli Cappelli, and L. Calamari. 2015. Effect of summer season on milk protein fractions in Holstein cows. J. Dairy Sci. 98:1815–1827. https://doi.org/10.3168/jds.2014-8788.
Bradford, B. J., K. Yuan, J. K. Farney, L. K. Mamedova, and A. J. Carpenter. 2015. Invited review: Inflammation during the transition to lactation: New adventures with an old flame. J. Dairy Sci. 98:6631–6650. https://doi.org/10.3168/jds.2015-9683.
Burhans, W. S., C. A. Rossiter Burhans, and L. H. Baumgard. 2022. Invited review: Lethal heat stress: The putative pathophysiology of a deadly disorder in dairy cattle. J. Dairy Sci. 105:3716–3735. https://doi.org/10.3168/jds.2021-21080.
Dahl, G. E., S. Tao, and J. Laporta. 2020. Heat stress impacts immune status in cows across the life cycle. Front. Vet. Sci. 7:116. https://doi.org/10.3389/fvets.2020.00116.
De Rensis, F., I. Garcia-Ispierto, and F. López-Gatius. 2015. Seasonal heat stress: Clinical implications and hormone treatments for the fertility of dairy cows. Theriogenology 84:659–666. https://doi.org/10.1016/j.theriogenology.2015.04.021.
Jensen, G. S., K. M. Patterson, and I. Yoon. 2008. Yeast culture has anti-inflammatory effects and specifically activates NK cells. Comp. Immunol. Microbiol. Infect. Dis. 31:487–500. https://doi.org/10.1016/j.cimid.2007.08.005.
Knoblock, C. E., W. Shi, I. Yoon, and M. Oba. 2019. Effects of supplementing a *Saccharomyces cerevisiae* fermentation product during the periparturient period on the immune response of dairy cows fed fresh diets differing in starch content. J. Dairy Sci. 102:6199–6209. https://doi.org/10.3168/jds.2018-16224.
Kononoff, P. J., A. J. Heinrichs, and D. R. Buckmaster. 2003. Modification of Penn State forage and total mixed ration particle separator and the effects of moisture content on its measurements. J. Dairy Sci. 86:1858–1863. https://doi.org/10.3168/jds.S0022-0302(03)73773-4.
Mahmoud, A. H. A., J. R. Slate, S. Hong, I. Yoon, and J. L. McGill. 2020. Supplementing a *Saccharomyces cerevisiae* fermentation product modulates innate immune function and ameliorates bovine respiratory syncytial virus infection in neonatal calves. J. Anim. Sci. 98:skaa252. https://doi.org/10.1093/jas/skaa252.
McCarthy, M. M., T. Yasui, M. J. B. Felippe, and T. R. Overton. 2016. Associations between the degree of early lactation inflammation and performance, metabolism, and immune function in dairy cows. J. Dairy Sci. 99:680–700. https://doi.org/10.3168/jds.2015-9694.
McGlone, J., S. Ford, F. Mitloehner, T. Grandin, P. Ruegg, C. Stull, G. Lewis, J. Swanson, W. Underwood, J. Mench, T. Mader, S. Eicher, P. Hester, J. Salak-Johnson, and M. Galyean. 2010. Dairy Cattle. Pages 74–89 in Guide for the Care and Use of Agriculture Animals in Research and Teaching. 3rd rev. ed. Federation of Animal Science Societies.
Menta, P. R., V. S. Machado, J. M. Piñeiro, W. W. Thatcher, J. E. P. Santos, and A. Vieira-Neto. 2022. Heat stress during the transition period is associated with impaired production, reproduction, and survival in dairy cows. J. Dairy Sci. 105:4474–4489. https://doi.org/10.3168/jds.2021-21185.
Micro-Tracers. Quality Assurance with Microtracer F. Accessed Mar. 31, 2022. https://microtracers.com/wp-content/uploads/2020/04/A-1-Quality-Assurance-with-Microtracer-F-6-12-13-ZE.pdf.
Miller-Webster, T., W. H. Hoover, M. Holt, and J. E. Nocek. 2002. Influence of yeast culture on ruminal microbial metabolism in continuous culture. J. Dairy Sci. 85:2009–2014. https://doi.org/10.3168/jds.S0022-0302(02)74277-X.
NRC. 1971. A Guide to Environmental Research on Animals. NRC.
NRC. 2001. Nutrient Requirements of Dairy Cattle. The National Academies Press.
Olagaray, K. E., S. E. Sivinski, B. A. Saylor, L. K. Mamedova, J. A. Sauls-Hiesterman, I. Yoon, and B. J. Bradford. 2019. Effect of *Saccharomyces cerevisiae* fermentation product on feed intake parameters, lactation performance, and metabolism of transition dairy cattle. J. Dairy Sci. 102:8092–8107. https://doi.org/10.3168/jds.2019-16315.
Orth, R. 1992. Sample Day and Lactation Report FactSheet A-2 DHIA 200. Mid-States Dairy Records Processing Center.
Polsky, L., and M. A. G. von Keyserlingk. 2017. Invited review: Effects of heat stress on dairy cattle welfare. J. Dairy Sci. 100:8645–8657. https://doi.org/10.3168/jds.2017-12651.
Poppy, G. D., A. R. Rabiee, I. J. Lean, W. K. Sanchez, K. L. Dorton, and P. S. Morley. 2012. A meta-analysis of the effects of feeding yeast culture produced by anaerobic fermentation of *Saccharomyces cerevisiae* on milk production of lactating dairy cows. J. Dairy Sci. 95:6027–6041. https://doi.org/10.3168/jds.2012-5577.
Robinson, P. H., and D. Meyer. 2010. Total Mixed Ration (TMR) Sampling Protocol. University of California. Agriculture and Natural Resources Publication 8413. ANR Communication Services.
Salminen, S., M. C. Collado, A. Endo, C. Hill, S. Lebeer, E. M. M. Quigley, M. E. Sanders, R. Shamir, J. R. Swann, H. Szajewska, and G. Vinderola. 2021. The International Scientific Association of Probiotics and Prebiotics (ISAPP) consensus statement on the definition and scope of postbiotics. Nat. Rev. Gastroenterol. Hepatol. 18:649–667. https://doi.org/10.1038/s41575-021-00440-6.
Sanz-Fernandez, M. V., S. K. Stoakes, J. S. Johnson, M. Abuajamieh, J. T. Seibert, S. C. Pearce, N. K. Gabler, R. P. Rhoads, and L. H. Baumgard. 2015. Heat stress; What's the gut got to do with it? Pages 42–57 in Proceedings of the 29th Annual Southwest Nutrition Conference.
Sargeant, J. M., A. M. O'Connor, I. A. Gardner, J. S. Dickson, and M. E. Torrence. 2010. The REFLECT statement: Reporting guidelines for randomized controlled trials in livestock and food safety: Explanation and elaboration. J. Food Prot. 73:579–603. https://doi.org/10.4315/0362-028X-73.3.579.
Saxton, A. M. 1998. A macro for converting mean separation output to letter groupings in Proc Mixed. Pages 1243–1246 in Proc. 23rd SAS Users Group Intl. SAS Institute Inc.
Schingoethe, D. J., K. N. Linke, K. F. Kalscheur, A. R. Hippen, D. R. Rennich, and I. Yoon. 2004. Feed efficiency of mid-lactation dairy cows fed yeast culture during summer. J. Dairy Sci. 87:4178–4181. https://doi.org/10.3168/jds.S0022-0302(04)73561-4.
Shi, W., C. E. Knoblock, I. Yoon, and M. Oba. 2019. Effects of supplementing a *Saccharomyces cerevisiae* fermentation product during the transition period on rumen fermentation of dairy cows fed fresh diets differing in starch content. J. Dairy Sci. 102:9943–9955. https://doi.org/10.3168/jds.2019-16671.
St-Pierre, N. R. 2007. Design and analysis of pen studies in the animal sciences. J. Dairy Sci. 90:E87–E99. https://doi.org/10.3168/jds.2006-612.
Tempelman, R. J. 2009. Invited review: Assessing experimental designs for research conducted on commercial dairies. J. Dairy Sci. 92:1–15. https://doi.org/10.3168/jds.2008-1404.
Thrune, M., A. Bach, M. Ruiz-Moreno, M. D. Stern, and J. G. Linn. 2009. Effects of *Saccharomyces cerevisiae* on ruminal pH and microbial fermentation in dairy cows. Livest. Sci. 124:261–265. https://doi.org/10.1016/j.livsci.2009.02.007.
Tun, H. M., S. Li, I. Yoon, S. J. Meale, P. A. Azevedo, E. Khafipour, and J. C. Plaizier. 2020. *Saccharomyces cerevisiae* fermentation products (SCFP) stabilize the ruminal microbiota of lactating dairy cows during periods of a depressed rumen pH. BMC Vet. Res. 16:237. https://doi.org/10.1186/s12917-020-02437-w.
Van Amburgh, M. E., E. A. Collao-Saenz, R. J. Higgs, D. A. Ross, E. B. Recktenwald, E. Raffrenato, L. E. Chase, T. R. Overton, J. K. Mills, and A. Foskolos. 2015. The Cornell Net Carbohydrate and Protein System: Updates to the model and evaluation of version 6.5. J. Dairy Sci. 98:6361–6380. https://doi.org/10.3168/jds.2015-9378.

Wheelock, J. B., R. P. Rhoads, M. J. VanBaale, S. R. Sanders, and L. H. Baumgard. 2010. Effects of heat stress on energetic metabolism in lactating Holstein cows. J. Dairy Sci. 93:644–655. https://doi.org/10.3168/jds.2009-2295.

Zhu, W., B. X. Zhang, K. Y. Yao, I. Yoon, Y. H. Chung, J. K. Wang, and J. X. Liu. 2016. Effects of supplemental levels of *Saccharomyces cerevisiae* fermentation product on lactation performance in dairy cows under heat stress. Asian-Australas. J. Anim. Sci. 29:801–806. https://doi.org/10.5713/ajas.15.0440.

Zontini, A. M., E. Zerbini, A. Minuti, and E. Trevisi. 2021. Effects of supplementing *Saccharomyces cerevisiae* fermentation products to dairy cows from the day of dry-off through early lactation. J. Dairy Sci. 104:11673–11685. https://doi.org/10.3168/jds.2021-20463.